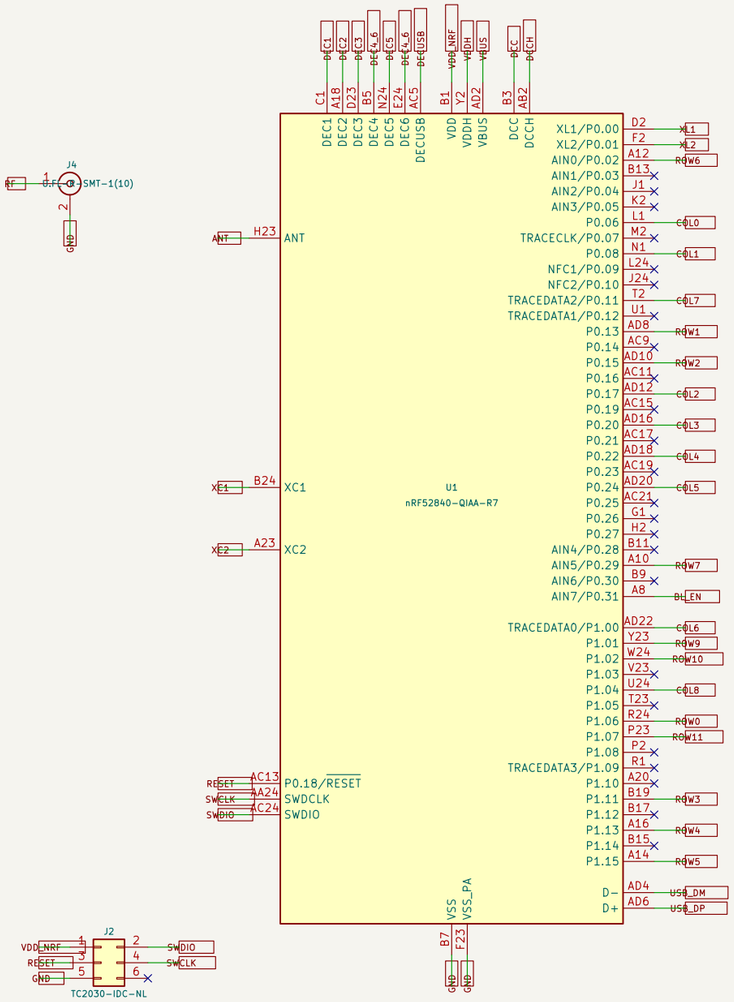
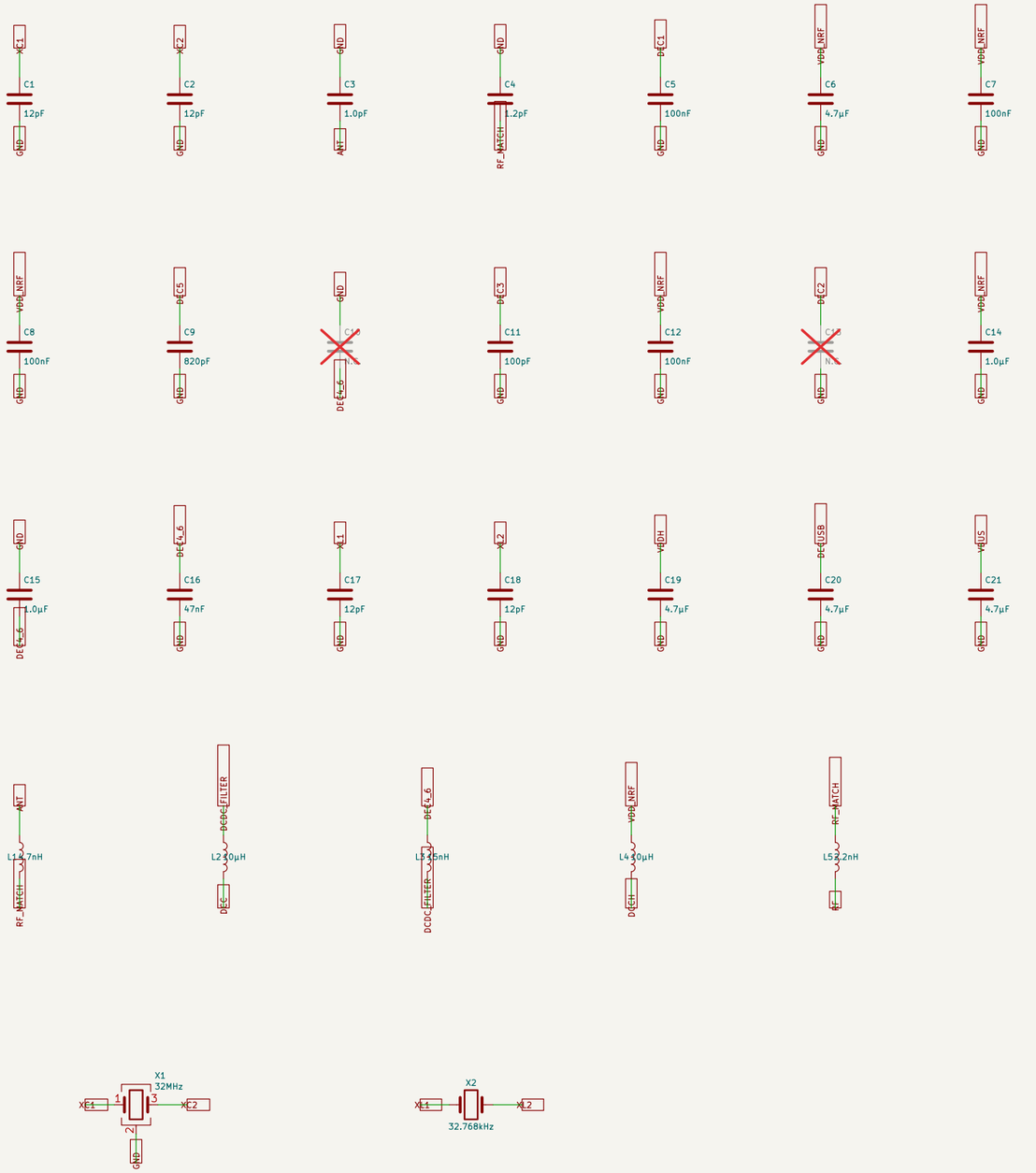




Matrix

File: matrix.kicad_sch

Power

File: power.kicad_sch

Nordic nRF52840-QIAA configuration 7
DEC3 is independent; DEC4 and DEC6 share DEC4_6.
C10 / C13: not fitted. C9: omit only for Fxx and later.
VDD_NRF / REGOUT0: 1.8 V (factory default).
ZMK: disable backlight before System OFF.

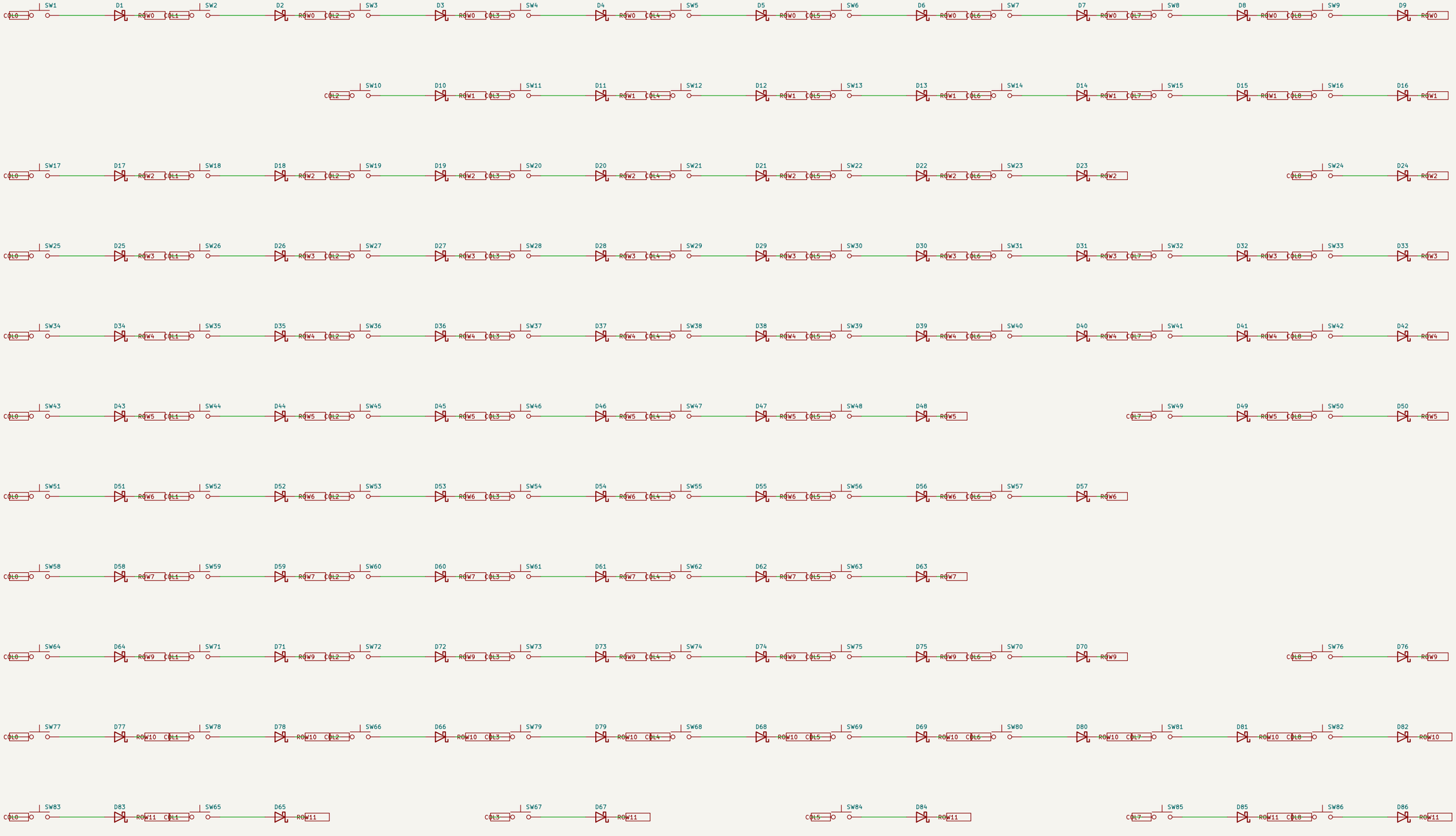
nRF52840 configuration 7; top LEDs / bottom electronics

Sheet: /
File: keyboard.kicad_sch

Title: TKL v8.2 – Controller

Size: A3	Date:	**Rev: 8.1**
KiCad E.D.A. 10.0.6		Id: 1/3

86 switches; cathodes toward ROW. ROW8 unused. D1–D86: BAT46W for 1.8 V GPIOs.

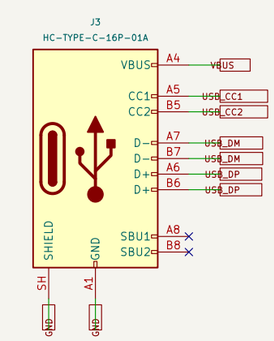


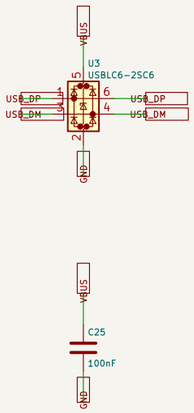
nRF52840 configuration 7; top LEDs / bottom electronics

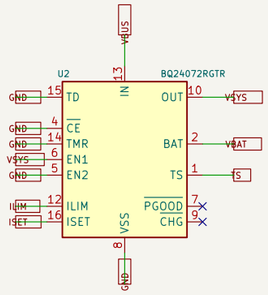
Sheet: /Matrix/
File: matrix.kicad_sch

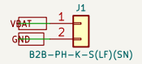
Title: TKL v8.2 – Matrix

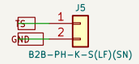
Size: A3	Date:	**Rev: 8.1**
KiCad E.D.A. 10.0.6		Id: 2/3

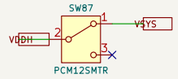


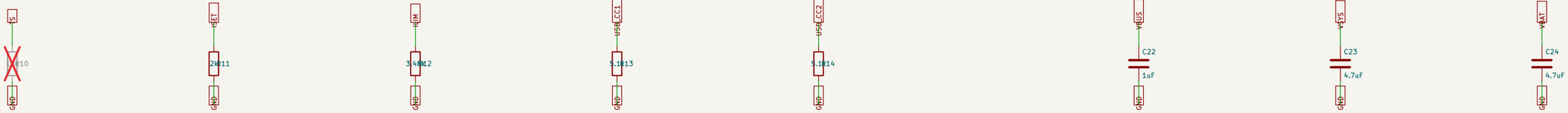


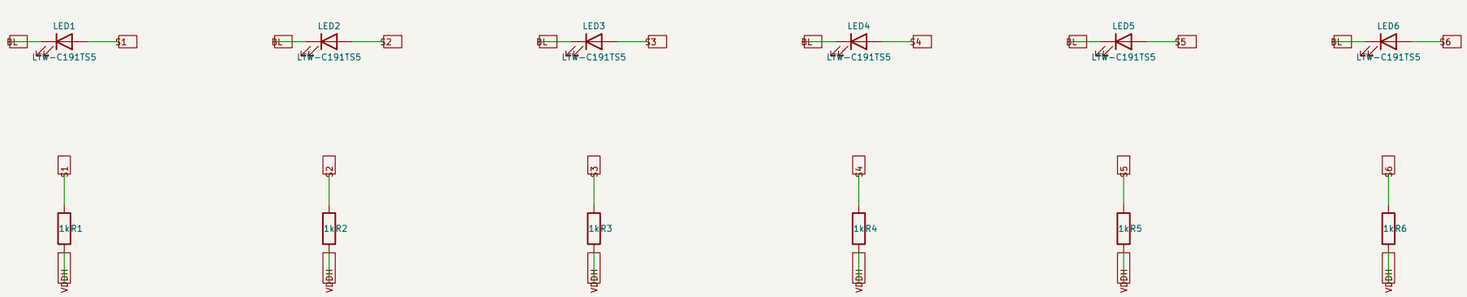


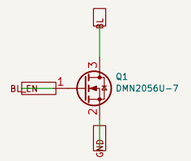














BQ24072: EN1=VSYS, EN2=0: USB500. R11=2k: ~445 mA charge.
TMR=0: timers disabled for 10 Ah pack. TD=0: termination enabled.
J5: external 10k NTC (103AT-2) on cell; R10 DNP. Protected 1S / 4.2 V pack.
Switch disconnects load; charging remains active. Use a 5 V / >=500 mA source.
Backlight: six top-emitting white LEDs from VDDH; 1k each, PWM via BL_EN.

nRF52840 configuration 7; top LEDs / bottom electronics

Sheet: /Power/
File: power.kicad_sch

Title: TKL v8.2 – Power

Size: A3	Date:	**Rev: 8.1**
KiCad E.D.A. 10.0.6		Id: 3/3